Final_Project_Malekshahi

November 21, 2025

0.1 Yoga and Mental Health

- An Exploratory Analysis
- Shaghayegh Malekshahi
- Data Exploration and Analysis

[16]:
```
# STEP 1: Import Libraries

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_rel
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import classification_report, confusion_matrix
import statsmodels.api as sm
from sklearn.metrics import silhouette_score
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
```

[2]:
```
# STEP 2: Load Datasets

# Load all three datasets
google_trends = pd.read_csv("google_trends_yoga.csv")
happiness_data = pd.read_csv("happiness_data.csv")
survey_data = pd.read_csv("survey.csv")
```

[3]:
```
# STEP 2A: Clean Google Trends Data
# Rename the 4 existing columns
google_trends.columns = ['Yoga', 'MentalHealth', 'Stress', 'IsPartial']
google_trends.head()
```

[3]:

	Yoga	MentalHealth	Stress	IsPartial
0	84	54	52	False
1	80	55	52	False
2	80	60	56	False

```
3    85            60      53      False
4    79            61      54      False
```

[4]:
```python
# STEP 2: Load and Clean Happiness Data
happiness_data = pd.read_csv("happiness_data.csv")

# Skip metadata rows and reset the index
happiness_data = happiness_data.iloc[2:].reset_index(drop=True)

# Rename columns
happiness_data.columns = ['Overall Rank', 'Country', 'Life Evaluation']

# Clean country names (remove footnotes)
import re
happiness_data['Country'] = happiness_data['Country'].apply(lambda x: re.
 ↪sub(r'\[.*?\]', '', str(x)).strip())

# Standardize casing
happiness_data['Country'] = happiness_data['Country'].str.title()

# Convert to numeric
happiness_data['Life Evaluation'] = pd.to_numeric(happiness_data['Life␣
 ↪Evaluation'], errors='coerce')
happiness_data['Overall Rank'] = pd.to_numeric(happiness_data['Overall Rank'],␣
 ↪errors='coerce')
happiness_data.head()
```

[4]:

	Overall Rank	Country	Life Evaluation
0	1	Lithuania	7.795
1	2	Israel	7.667
2	3	Iceland	7.658
3	4	Serbia	7.658
4	5	Denmark	7.598

[7]:
```python
# STEP 2C: Clean Survey Data

# Rename columns for clarity
survey_data = survey_data.rename(columns={
    'self_employed': 'SelfEmployed',
    'family_history': 'FamilyHistory',
    'work_interfere': 'WorkInterfere',
    'no_employees': 'NumEmployees',
    'remote_work': 'RemoteWork',
    'tech_company': 'TechCompany',
    'wellness_program': 'WellnessProgram',
    'seek_help': 'SeekHelp',
    'mental_health_consequence': 'MentalHealthConsequence',
```

```
    'phys_health_consequence': 'PhysicalHealthConsequence',
    'mental_health_interview': 'MentalHealthInterview',
    'phys_health_interview': 'PhysicalHealthInterview',
    'mental_vs_physical': 'MentalVsPhysical',
    'obs_consequence': 'ObservedConsequence'
})

# Standardize 'Gender' values
survey_data['Gender'] = survey_data['Gender'].str.strip().str.lower()
survey_data['Gender'] = survey_data['Gender'].replace({
    'female': 'Female', 'f': 'Female', 'cis female': 'Female', 'female (cis)':␣
 ↪'Female',
    'male': 'Male', 'm': 'Male', 'cis male': 'Male', 'male-ish': 'Male',
    'trans-female': 'Transgender', 'trans male': 'Transgender',
    'non-binary': 'Non-binary', 'genderqueer': 'Non-binary',
    'agender': 'Non-binary', 'other': 'Other', 'fluid': 'Non-binary'
})

# Drop rows with 3 or more missing values
survey_data = survey_data.dropna(thresh=survey_data.shape[1] - 3)

# Fill missing values in SelfEmployed
survey_data['SelfEmployed'] = survey_data['SelfEmployed'].fillna('Unknown')
survey_data.head()
```

[7]:

```
             Timestamp  Age  Gender         Country state SelfEmployed  \
0  2014-08-27 11:29:31   37  Female   United States    IL      Unknown
1  2014-08-27 11:29:37   44    Male   United States    IN      Unknown
2  2014-08-27 11:29:44   32    Male          Canada   NaN      Unknown
3  2014-08-27 11:29:46   31    Male  United Kingdom   NaN      Unknown
4  2014-08-27 11:30:22   31    Male   United States    TX      Unknown

  FamilyHistory treatment WorkInterfere    NumEmployees  …  \
0            No       Yes         Often            6-25  …
1            No        No        Rarely  More than 1000  …
2            No        No        Rarely            6-25  …
3           Yes       Yes         Often          26-100  …
4            No        No         Never         100-500  …

                leave MentalHealthConsequence PhysicalHealthConsequence  \
0       Somewhat easy                      No                        No
1          Don't know                   Maybe                        No
2  Somewhat difficult                      No                        No
3  Somewhat difficult                     Yes                       Yes
4          Don't know                      No                        No

      coworkers supervisor MentalHealthInterview PhysicalHealthInterview  \
```

```
0  Some of them        Yes                    No                    Maybe
1            No         No                    No                       No
2           Yes        Yes                   Yes                      Yes
3  Some of them         No                 Maybe                    Maybe
4  Some of them        Yes                   Yes                      Yes

  MentalVsPhysical ObservedConsequence comments
0              Yes                  No      NaN
1       Don't know                  No      NaN
2               No                  No      NaN
3               No                 Yes      NaN
4       Don't know                  No      NaN

[5 rows x 27 columns]
```

[8]:
```
# STEP 3: Add Date Column to Google Trends Data

# Since no date column existed, we simulate one starting from Jan 1, 2023␣
↪(weekly data)
google_trends['Date'] = pd.date_range(start='2023-01-01',␣
↪periods=len(google_trends), freq='W')
google_trends.head()
```

[8]:

	Yoga	MentalHealth	Stress	IsPartial	Date
0	84	54	52	False	2023-01-01
1	80	55	52	False	2023-01-08
2	80	60	56	False	2023-01-15
3	85	60	53	False	2023-01-22
4	79	61	54	False	2023-01-29

[9]:
```
print(google_trends.columns.tolist())
```

```
['Yoga', 'MentalHealth', 'Stress', 'IsPartial', 'Date']
```

[10]:
```
# STEP 4: Summary Statistics of Key Variables

# Descriptive stats for univariate analysis
print("Summary Statistics:\n")
google_trends[['Yoga', 'MentalHealth', 'Stress']].describe()
```

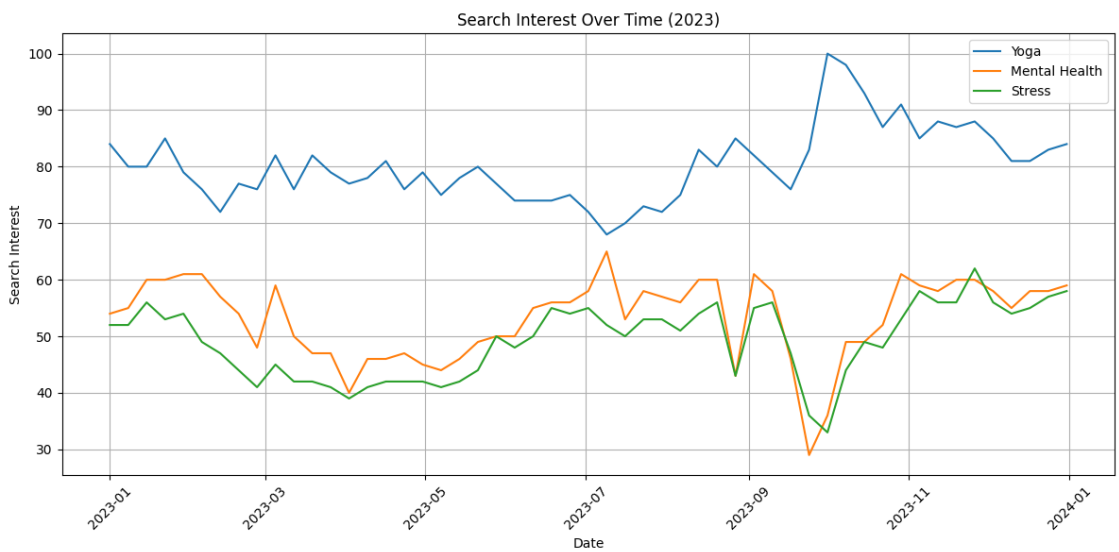
```
Summary Statistics:
```

[10]:

	Yoga	MentalHealth	Stress
count	53.000000	53.000000	53.000000
mean	80.283019	53.188679	49.207547
std	6.499637	7.248492	6.564352
min	68.000000	29.000000	33.000000

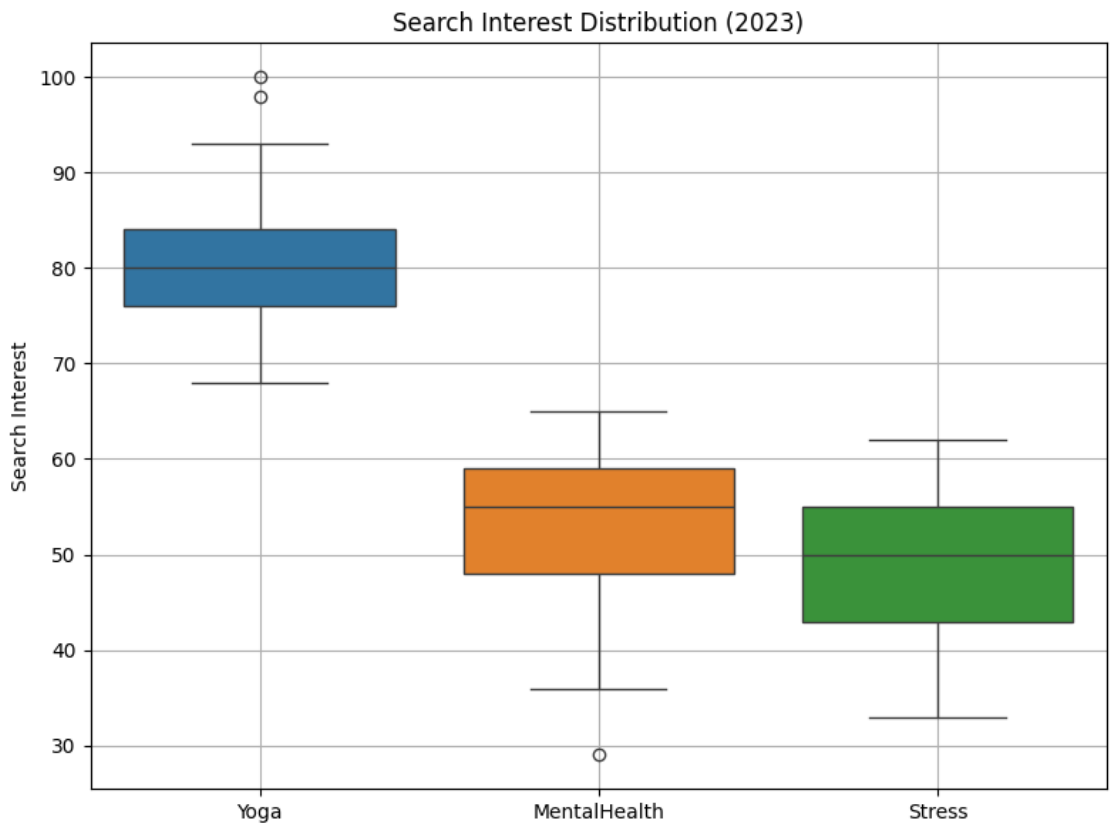
```
25%      76.000000     48.000000  43.000000
50%      80.000000     55.000000  50.000000
75%      84.000000     59.000000  55.000000
max     100.000000     65.000000  62.000000
```

[11]:
```
# STEP 5: Visualization of Key Variables

# Line plot showing trends over time
plt.figure(figsize=(12, 6))
plt.plot(google_trends['Date'], google_trends['Yoga'], label='Yoga')
plt.plot(google_trends['Date'], google_trends['MentalHealth'], label='Mental␣
↪Health')
plt.plot(google_trends['Date'], google_trends['Stress'], label='Stress')
plt.title('Search Interest Over Time (2023)')
plt.xlabel('Date')
plt.ylabel('Search Interest')
plt.legend()
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
```



[12]:
```
# Boxplot for distribution
plt.figure(figsize=(8, 6))
sns.boxplot(data=google_trends[['Yoga', 'MentalHealth', 'Stress']])
plt.title('Search Interest Distribution (2023)')
plt.ylabel('Search Interest')
plt.grid(True)
```

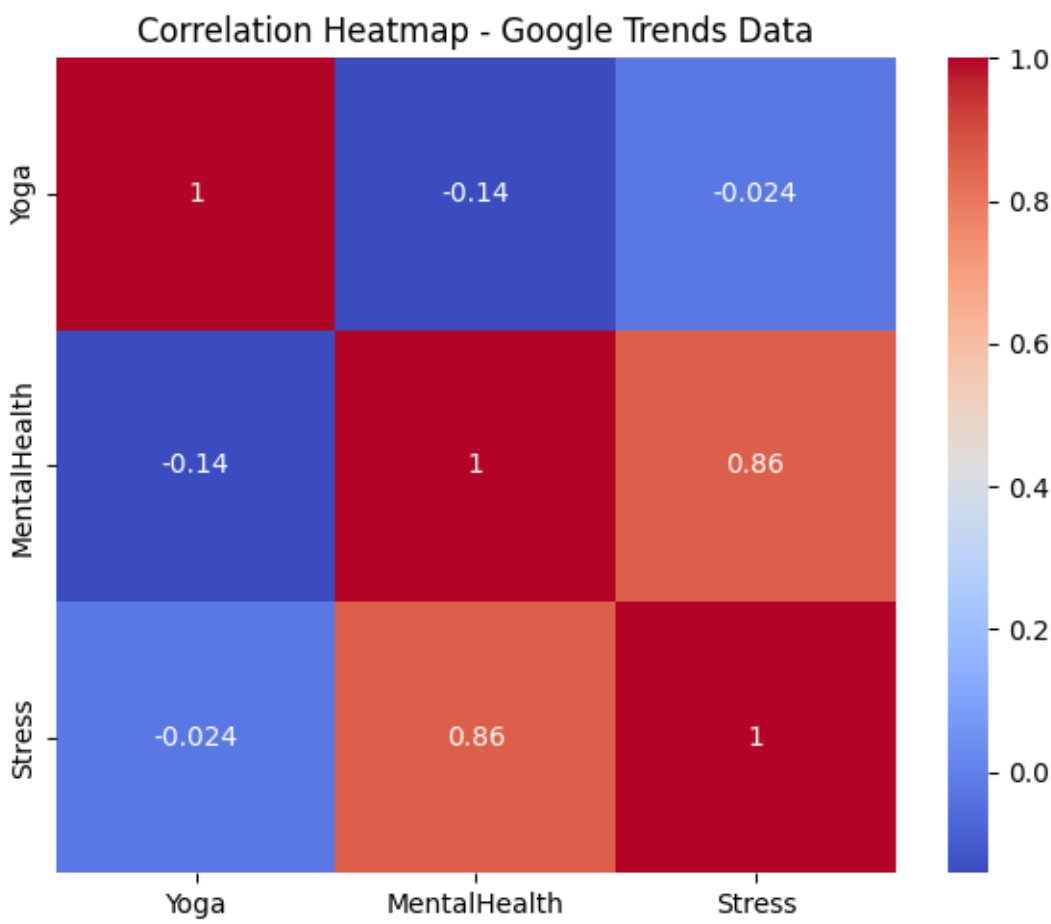
```
plt.tight_layout()
plt.show()
```

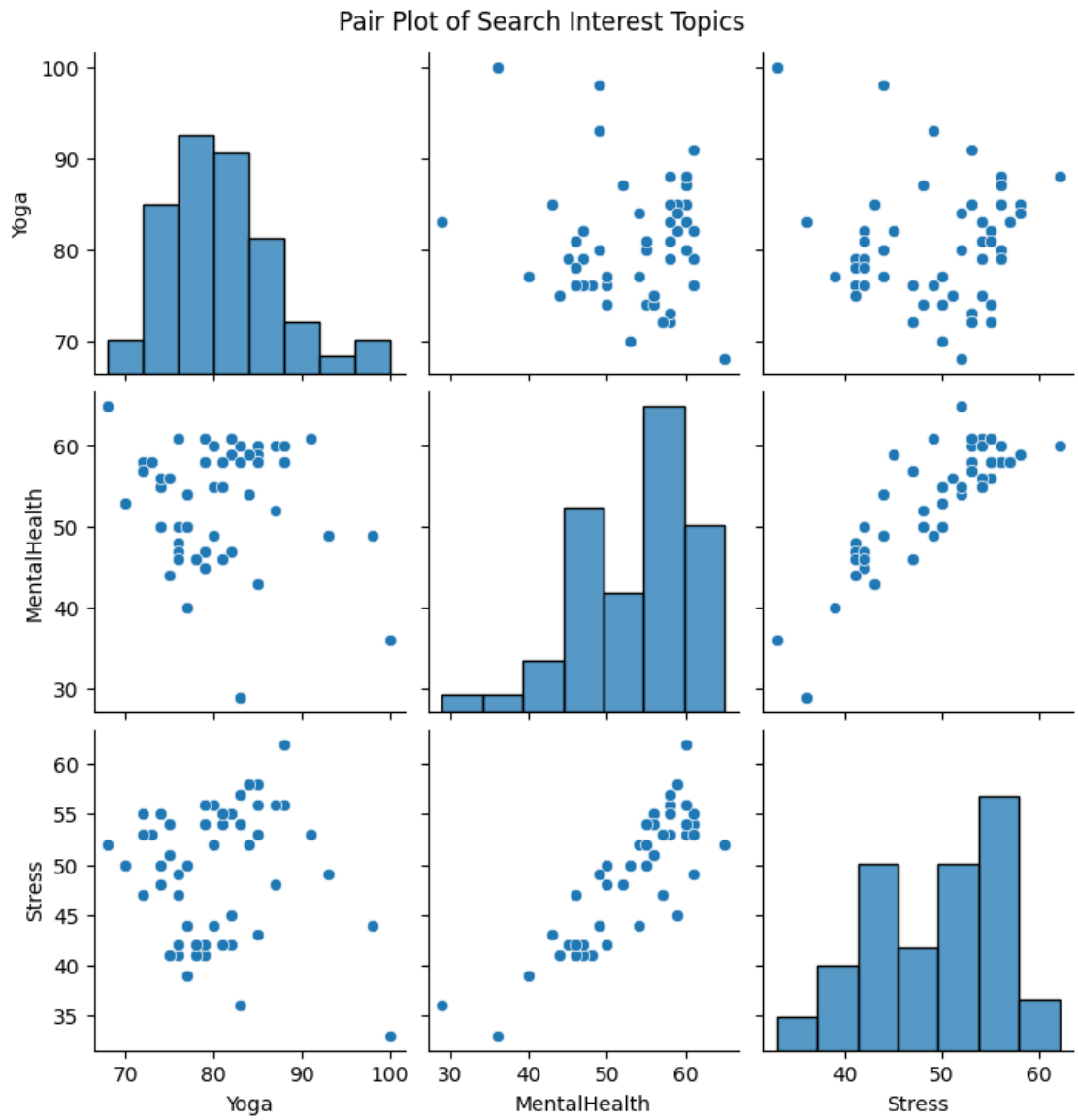


[13]:
```
# STEP 6: Bivariate Analysis (Covariance & Correlation)

# Covariance matrix
print("\nCovariance Matrix:\n")
print(google_trends[['Yoga', 'MentalHealth', 'Stress']].cov())

# Pearson correlation matrix
print("\nCorrelation Matrix:\n")
print(google_trends[['Yoga', 'MentalHealth', 'Stress']].corr(method='pearson'))

# Heatmap of correlation
plt.figure(figsize=(6, 5))
sns.heatmap(google_trends[['Yoga', 'MentalHealth', 'Stress']].corr(),␣
↪annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap - Google Trends Data')
plt.tight_layout()
plt.show()
```

```python
# Pair plot
sns.pairplot(google_trends[['Yoga', 'MentalHealth', 'Stress']])
plt.suptitle('Pair Plot of Search Interest Topics', y=1.02)
plt.show()
```

```
Covariance Matrix:

                   Yoga  MentalHealth     Stress
Yoga          42.245283     -6.669811  -1.021408
MentalHealth  -6.669811     52.540639  40.979318
Stress        -1.021408     40.979318  43.090711

Correlation Matrix:

                   Yoga  MentalHealth    Stress
Yoga          1.000000     -0.141572 -0.023940
MentalHealth -0.141572      1.000000  0.861242
Stress       -0.023940      0.861242  1.000000
```





[14]:
```
# STEP 7: Hypothesis Testing (Paired t-test)

# H0: Yoga interest is <= mental health interest
# H1: Yoga interest is > mental health interest
# Hypothesis Test: Is Yoga > Mental Health in average weekly search interest?
t_stat, p_val = ttest_rel(google_trends['Yoga'], google_trends['MentalHealth'],␣
↪alternative='greater')

print("\nPaired t-Test Result:")
```

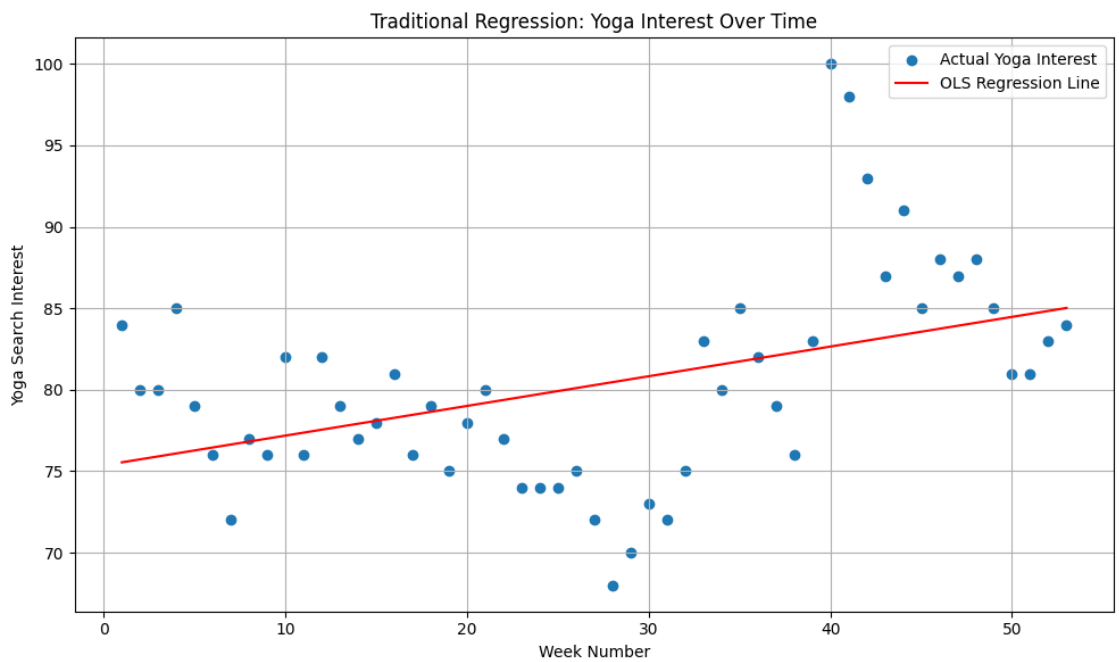
```python
print(f"t-statistic = {t_stat:.4f}, p-value = {p_val:.4e}")
```

```
Paired t-Test Result:
t-statistic = 18.9693, p-value = 2.5248e-25
```

[15]:
```python
# STEP 8: Traditional Linear Regression using statsmodels

# Create week index variable
google_trends['week_num'] = range(1, len(google_trends) + 1)
X = sm.add_constant(google_trends[['week_num']])
y = google_trends['Yoga']
model = sm.OLS(y, X).fit()

# Print model summary
print(model.summary())


# Plot regression line
plt.figure(figsize=(10, 6))
plt.scatter(google_trends['week_num'], y, label='Actual Yoga Interest')
plt.plot(google_trends['week_num'], model.predict(X), color='red', label='OLS␣
↪Regression Line')
plt.xlabel('Week Number')
plt.ylabel('Yoga Search Interest')
plt.title('Traditional Regression: Yoga Interest Over Time')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   Yoga   R-squared:                       0.187
Model:                            OLS   Adj. R-squared:                  0.171
Method:                 Least Squares   F-statistic:                     11.75
Date:                Tue, 04 Nov 2025   Prob (F-statistic):            0.00121
Time:                        18:34:13   Log-Likelihood:                -168.41
No. Observations:                  53   AIC:                             340.8
Df Residuals:                      51   BIC:                             344.8
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         75.3650      1.649     45.712      0.000      72.055      78.675
week_num       0.1821      0.053      3.428      0.001       0.075       0.289
==============================================================================
```

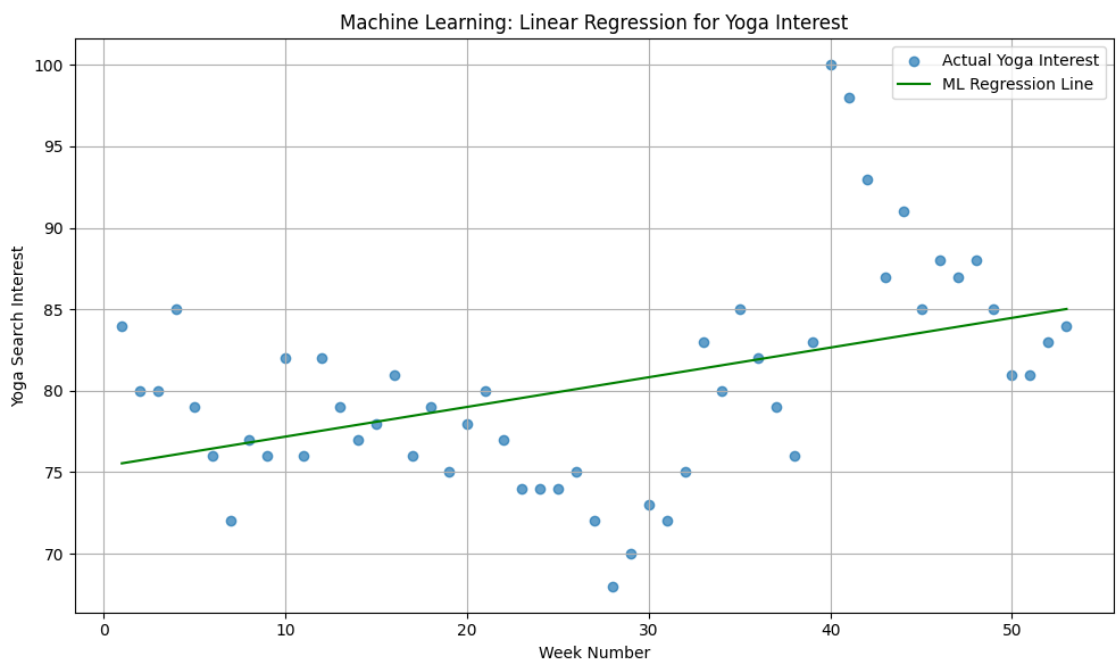
```
Omnibus:                        5.256   Durbin-Watson:                   0.567
Prob(Omnibus):                  0.072   Jarque-Bera (JB):                4.221
Skew:                           0.548   Prob(JB):                        0.121
Kurtosis:                       3.842   Cond. No.                         63.0
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
```



[17]:
```python
# STEP 9: Machine Learning Regression using scikit-learn

ml_model = LinearRegression()
X_ml = google_trends[['week_num']]
y_ml = google_trends['Yoga']
ml_model.fit(X_ml, y_ml)
y_pred = ml_model.predict(X_ml)

# Evaluate model
mse = mean_squared_error(y_ml, y_pred)
r2 = r2_score(y_ml, y_pred)
print("\nMachine Learning Linear Regression:")
print(f"Intercept = {ml_model.intercept_:.2f}")
print(f"Coefficient = {ml_model.coef_[0]:.4f}")
print(f"MSE = {mse:.2f}")
```

```
print(f"R² Score = {r2:.3f}")

# Plot ML regression
plt.figure(figsize=(10, 6))
plt.scatter(X_ml, y_ml, label='Actual Yoga Interest', alpha=0.7)
plt.plot(X_ml, y_pred, color='green', label='ML Regression Line')
plt.xlabel('Week Number')
plt.ylabel('Yoga Search Interest')
plt.title('Machine Learning: Linear Regression for Yoga Interest')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

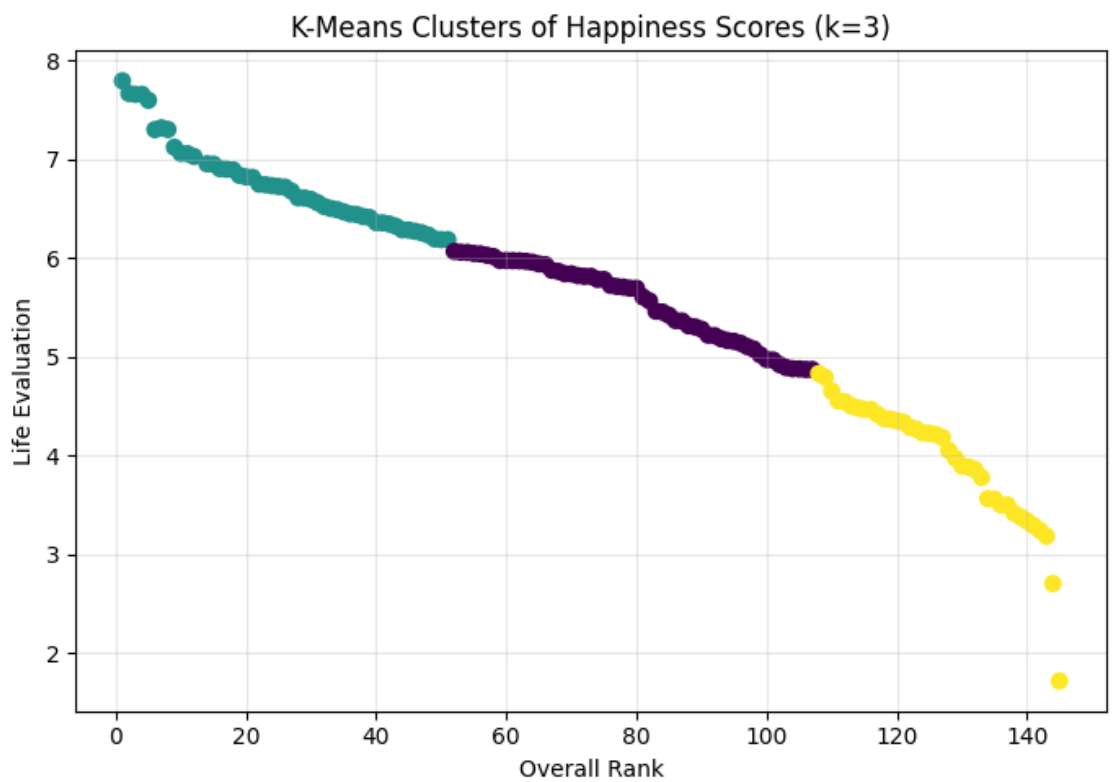
```
Machine Learning Linear Regression:
Intercept = 75.37
Coefficient = 0.1821
MSE = 33.68
R² Score = 0.187
```



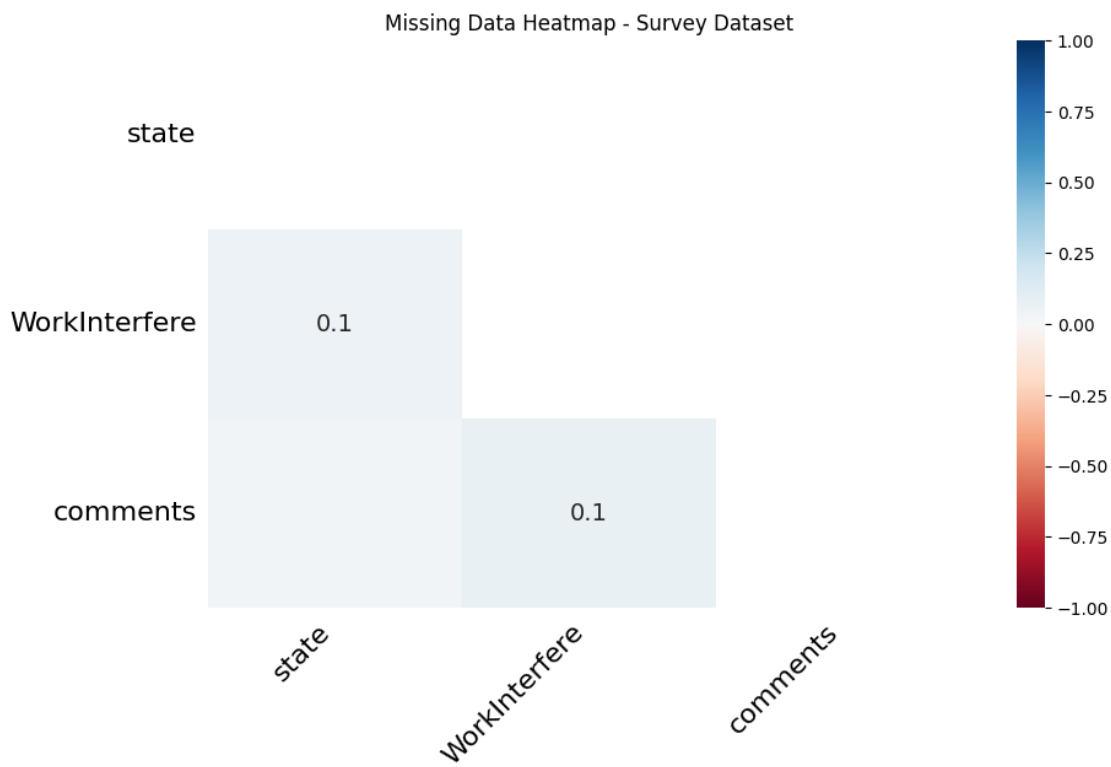
[18]:
```
# STEP 10: K-Means Clustering on Happiness Data
hdf = happiness_data.copy()
hdf = hdf.dropna(subset=['Life Evaluation'])  # ensure valid rows

# Standardize numeric columns for clustering
```

```python
features = ['Overall Rank', 'Life Evaluation']
X = StandardScaler().fit_transform(hdf[features])

# Run K-Means with k=3
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
labels = kmeans.fit_predict(X)
hdf['Cluster'] = labels

# Evaluate clustering
silhouette = silhouette_score(X, labels)
print(f"Silhouette Score (k=3): {silhouette:.3f}")

# Visualize clusters
plt.figure(figsize=(7,5))
plt.scatter(hdf['Overall Rank'], hdf['Life Evaluation'], c=hdf['Cluster'], s=45)
plt.xlabel('Overall Rank')
plt.ylabel('Life Evaluation')
plt.title('K-Means Clusters of Happiness Scores (k=3)')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

```
Silhouette Score (k=3): 0.551
```



[19]:
```
import missingno as msno

# Display missing value heatmap for the survey data
msno.heatmap(survey_data, figsize=(10, 6))
plt.title('Missing Data Heatmap - Survey Dataset')
plt.show()
```



0.1.1 Logistic Regression Code (Predicting Mental Health Treatment)

Classification Model: Logistic Regression

Predict if a person has received treatment

[20]:
```
# Import required libraries
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, confusion_matrix
```

[21]:
```
# Load the survey dataset (ensure the file is in your working directory)
survey_data = pd.read_csv("survey.csv")
```

[22]:
```
#  Select relevant columns for classification
# Target: treatment | Features: age, gender, family history, work interference
survey_class = survey_data[['Age', 'Gender', 'family_history',␣
↪'work_interfere', 'treatment']].copy()
```

[23]:
```
# Drop rows with missing values to simplify preprocessing
survey_class.dropna(inplace=True)
```

[24]:
```
# Encode categorical variables using LabelEncoder
le = LabelEncoder()
for col in ['Gender', 'family_history', 'work_interfere', 'treatment']:
    survey_class[col] = le.fit_transform(survey_class[col])
```

[25]:
```
# Split data into features (X) and target (y)
X_class = survey_class.drop('treatment', axis=1)
y_class = survey_class['treatment']
```

[26]:
```
# Split into training and test sets (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X_class, y_class,␣
↪test_size=0.2, random_state=42)
```

[27]:
```
# Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

[28]:
```
# Train a logistic regression model
logreg = LogisticRegression(max_iter=1000)
logreg.fit(X_train, y_train)
```

[28]:
```
LogisticRegression(max_iter=1000)
```

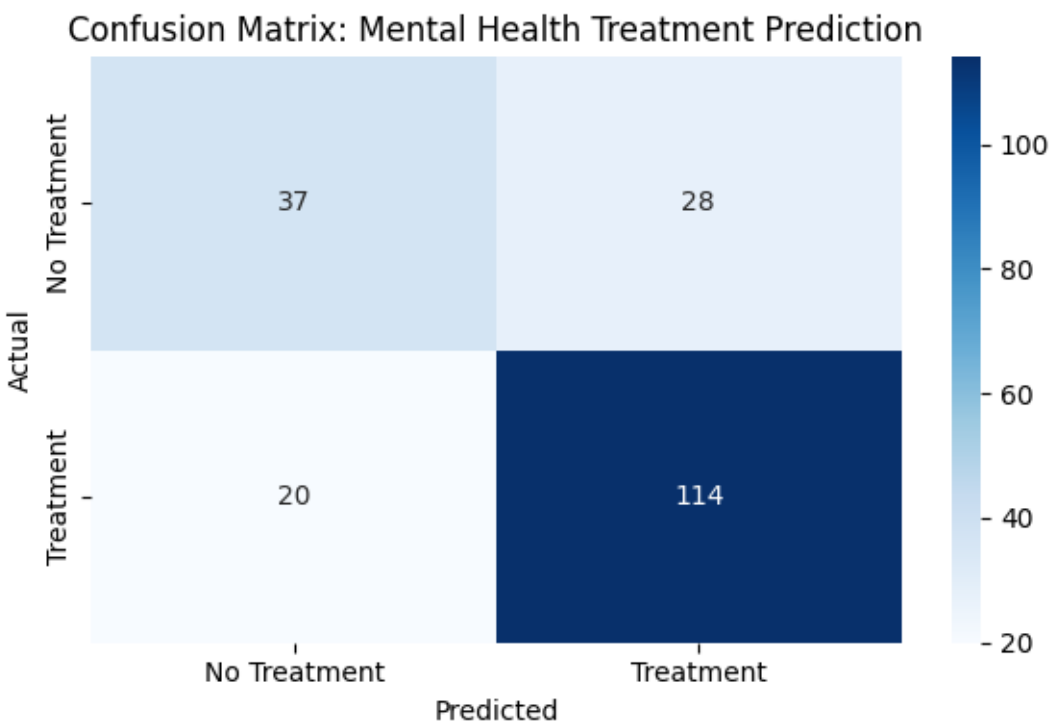
[29]:
```
# Make predictions on the test set
y_pred = logreg.predict(X_test)
```

[30]:
```
# Evaluate the model
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

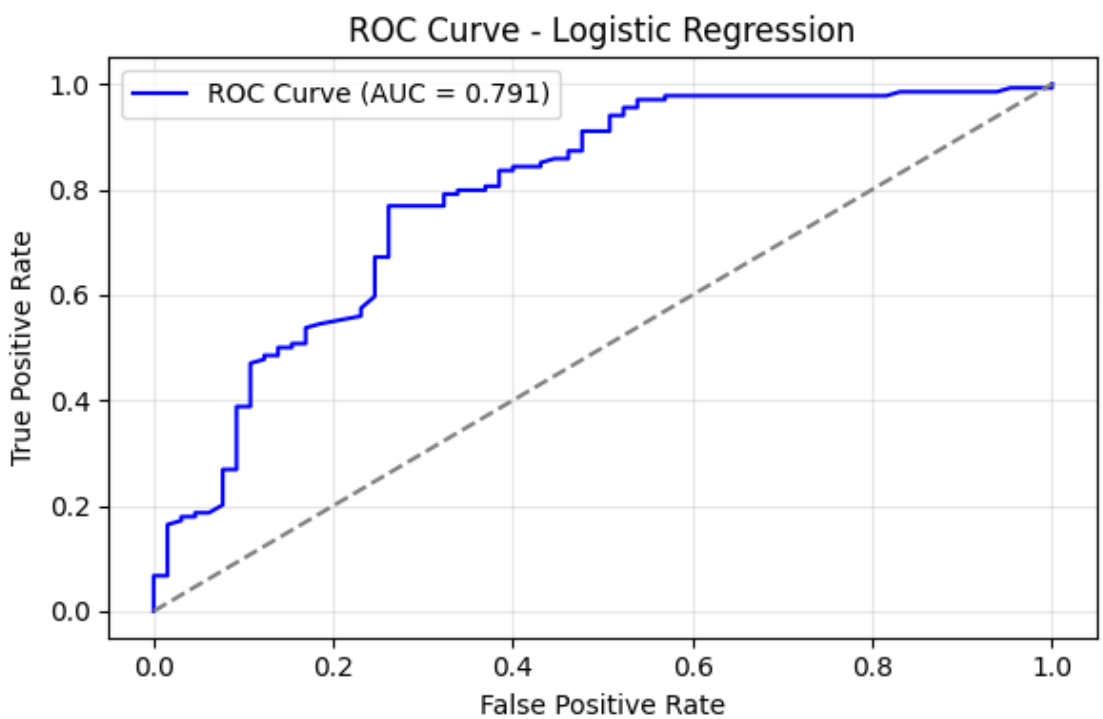
```
Confusion Matrix:
 [[ 37  28]
 [ 20 114]]

Classification Report:
               precision    recall  f1-score   support

           0       0.65      0.57      0.61        65
           1       0.80      0.85      0.83       134
```

```
    accuracy                           0.76       199
   macro avg       0.73      0.71      0.72       199
weighted avg       0.75      0.76      0.75       199
```

[31]:
```python
# Create and plot confusion matrix
cm = confusion_matrix(y_test, y_pred)
labels = ['No Treatment', 'Treatment']

plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=labels,␣
 ↪yticklabels=labels)
plt.title('Confusion Matrix: Mental Health Treatment Prediction')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.tight_layout()
plt.show()
```



[32]:
```python
# ---- Additional Evaluation Metrics ----
from sklearn.metrics import roc_auc_score, roc_curve

# Probability scores for ROC-AUC
```

```python
y_proba = logreg.predict_proba(X_test)[:, 1]
roc_auc = roc_auc_score(y_test, y_proba)

print("\nROC-AUC Score:", round(roc_auc, 3))

# Plot ROC curve
fpr, tpr, thresholds = roc_curve(y_test, y_proba)
plt.figure(figsize=(6, 4))
plt.plot(fpr, tpr, color='blue', label=f'ROC Curve (AUC = {roc_auc:.3f})')
plt.plot([0, 1], [0, 1], '--', color='gray')
plt.title('ROC Curve - Logistic Regression')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

```
ROC-AUC Score: 0.791
```